



Original Research

Assessment of Health Risks of Nitrate and Heavy Metal-Rich Groundwater: A Case Study of Baghpat City of Western Uttar Pradesh, India

Praveen Singh¹, Mukesh Ruhela^{1,*}  and Faheem Ahamad^{2,3} 

¹ Department of Environmental Engineering, Swami Vivekanand Subharti University, Meerut-250005, Uttar Pradesh, India

² Limnology and Ecological Modelling Laboratory, Department of Zoology and Environmental Sciences, Gurukul Kangri (Deemed to be University), Haridwar 249404, Uttarakhand, India

³ SAM (India) Builtwell Pvt. Ltd., Group VIII Project, IIM Indore, Madhya Pradesh, India

* Corresponding author. Email: mruhela@gmail.com



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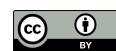
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Abstract

The study was performed to document the quality of groundwater in the agriculturally intensive Baghpat city of western Uttar Pradesh, India. A total of ten sites were selected strategically in the study area, combining agricultural fields, residential areas, commercial areas, and along the highway joining Meerut and Baghpat cities. A total of 240 groundwater samples were taken using the composite method of sampling and analysed for the selected parameters. The study revealed the values of total dissolved solids (TDS) and total hardness (TH) beyond the standard permissible limit of the Bureau of Indian Standards (BIS). The values of iron (Fe) were found above the standard permissible limit of BIS (0.3 mg/l) at all the sites except at GWS6 and 8. The values of zinc (Zn) were found within the standard permissible limit of BIS (5.0 mg/l). However, the values of the rest of the analysed heavy metals, i.e., arsenic (As), copper (Cu), lead (Pb), and cadmium (Cd), were found below the detectable limit (BDL). The health risk index (HRI) values were found below 1 for men, women, and children, indicating no risk to the health of human beings. Among men, women, and children, the highest risk was observed for children, followed by women and men. Based on the findings of the current study, it can be concluded that although the groundwater is safe for drinking with reference to heavy metals, the elevated values of TDS and TH necessitate proper treatment before it is used for drinking.

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Statement of Sustainability: The current study of groundwater quality assessment was carried out in an agriculture-intensive area, where the possibility of contamination is high due to the leaching of ions from fertilizers and pesticides. Along with this, irrigation practices using polluted river water are another major cause of groundwater contamination. The literature review revealed the lack of groundwater quality assessment studies focusing on health risk assessment. Therefore, the current study was carried out to assess the groundwater quality of agriculturally intensive Baghpat city of western Uttar Pradesh, India.

1. Introduction

Groundwater is one of the key and irreplaceable natural resources, essential for human survival on this planet. Currently, groundwater resources are under pressure with reference to their quality and quantity due to increasing urbanization, industrialization, population explosion, and greed for a lavish life. Additionally, the unjustified use of agrochemicals to enhance crop production is a major threat to groundwater quality (Verma et al., 2023; Etikala et al., 2024; Tripathi et al., 2025). The lack of skilled labour and wastewater treatment infrastructure aggravates the problem of groundwater pollution. The problem of groundwater pollution is in the worst condition in developing nations, which are struggling to provide even basic water supplies to their population (Ahamad et al., 2025). The polluted water transmits diseases like typhoid, diarrhoea, dysentery, cholera, and polio and is responsible for more than 500,000 deaths each year (UN, 2021).

In India, groundwater is more polluted in agricultural and industrial regions compared to other regions due to the release of agrochemicals and industrial contaminants. In northern India, the Indo-Gangetic plain is such an area where the problem of groundwa-



ter pollution is increasing continuously (Verma et al., 2023; Bojago et al., 2023). In the groundwater of agriculture-intensive areas, the contaminants frequently observed were nitrate, phosphate, sulphate, and heavy metals, which may be due to both geogenic processes and anthropogenic inputs (Duvva et al., 2021; Picetti et al., 2022). In the Indo-Gangetic plains, the major sources of nitrate pollution are agrochemicals, manures, leakage from open drains, and septic tanks (Adimalla & Qian, 2019; Verma et al., 2023; Ahamad et al., 2024).

The consumption of contaminated water is responsible for various diseases in human beings. The high values of nitrate and heavy metals in groundwater, when they enter the human body, cause various somatic and genetic changes, and in the long term result in several non-carcinogenic and carcinogenic risks (Adimalla & Qian, 2020; Maleki & Jari, 2021; Picetti et al., 2022; Ahamad et al., 2023; Ahamad et al., 2024). Children are generally found to be the most vulnerable to health risks, followed by women and men, which may be due to a higher hazard quotient and lower body weight in children and women (Chabukdhara et al., 2017; Zarei et al., 2022; Ayejoto & Egbueri, 2023).

In different districts of Uttar Pradesh, i.e., Meerut, Ghaziabad, Kanpur, Bareilly, Sonbhadra, and Mirzapur, groundwater is found unsafe for domestic uses due to values beyond the BIS standard limits of heavy metals (Chabukdhara et al., 2017; Ahamad et al., 2020; Mazhar & Ahmad, 2020; Gupta et al., 2022; Ruhela et al., 2024). In and around the districts of the National Capital Region of Delhi (Meerut, Ghaziabad, Baghpat), the population, consumption patterns, and site-specific conditions (land use, climate, seasons, and groundwater recharge patterns) are changing rapidly, necessitating continuous monitoring of groundwater to detect contamination and protect human health. However, few studies have been conducted in the Baghpat area of Western Uttar Pradesh exploring the groundwater quality and its health risks. Therefore, the current study was carried out to evaluate the groundwater quality of the Baghpat area with reference to physicochemical parameters and heavy metals. The health risk assessment was also carried out due to the ingestion of nitrate and heavy metal-rich groundwater.

2. Materials and Methods

2.1. Study Area

Ten distinct locations in the Baghpat area of Uttar Pradesh were chosen to achieve the study's goal. The district is bounded by Meerut to the east, Muzaffarnagar to the north, and Ghaziabad to the south. The majority of the people living in the district are engaged in agriculture. The district receives roughly 768 mm of rainfall on average each year and has a sub-humid to subtropical climate, with local temperatures ranging from 3 to 43 °C. All the selected clusters are shown in Figure 1.

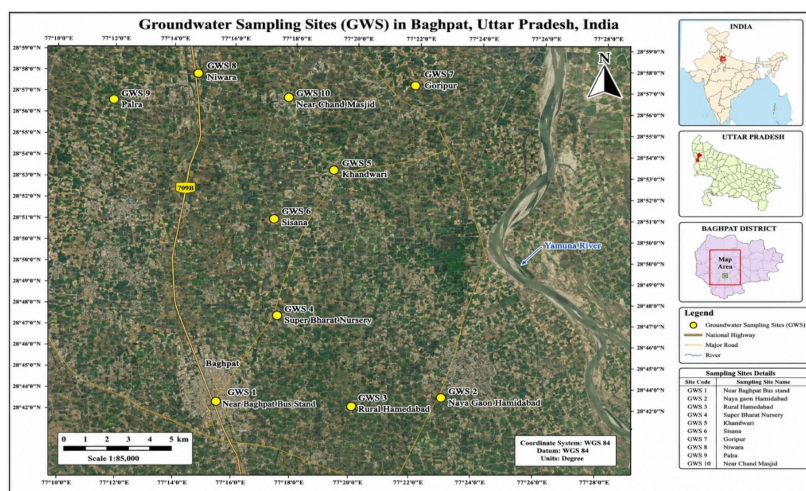


Figure 1. Distribution of selected groundwater sampling sites.

2.2. Sampling and Analysis

From June 2024 to May 2026, composite water samples were collected once every three months from each sampling cluster. Three samples were collected from each sampling cluster at each sampling time, and therefore, a total of 240 groundwater samples were collected throughout the study period from all the sampling sites. pH, TDS, TH, nitrate (NO_3^-), As, Fe, Cu, Zn, Pb, and Cd were all measured in the water samples following the standard procedures prescribed by the American Public Health Association (APHA, 2017). A digital meter (pHep HI98107) was used to measure pH at the sampling sites.



2.3. Health Risk Assessment

The term “human health risk assessment” was introduced by the US Environmental Protection Agency (USEPA, 1989) to characterize the possible adverse health effects of exposure to different contaminants through the consumption of contaminated groundwater (Li et al., 2016; Zhang et al., 2018). The ingestion pathway was taken into account in this study to assess the non-carcinogenic health risks of the chosen ions (zinc, iron, and nitrate). The following methods were used to calculate the non-carcinogenic health risk index (HRI) from the consumption of Fe, Zn, and NO₃⁻.

1. First, the chronic daily intake was calculated using the equation below:

$$\text{Chronic Daily Intake of ingestion } [CDI(i)] = \frac{(Cw \times IR \times EF \times ED)}{(BW \times AET)} \quad (1)$$

Where, Cw is the concentration of certain ions in water; IR = Ingestion rate (0.78 L/day for children and 2.5 L/day for men and women) (USEPA, 2014); EF = Exposure frequency (365 days per year) (USEPA, 2014); ED = Exposure duration (64 years for men, 67 years for women, and 12 years for children) (WHO, 2013; Narsimha & Rajitha, 2018); BW = Body weight (15 kg for children, 55 kg for women, and 65 kg for men) (ICMR, 2009); AET = Average exposure time (ED × 365 days);

2. Then, the hazard quotient (HQ_i) of ingestion was calculated using the equation below:

$$\text{Hazard Quotient of ingestion } (HQ_i) = \frac{CDI(i)}{RFD(i)} \quad (2)$$

where RFD is the oral reference dose; per USEPA (2014), the RFD for nitrate is 1.6 mg/kg/d, for Fe is 0.7 mg/kg/d, and for Zn is 0.3 mg/kg/d.

3. Finally, the total hazard index (THI), or health risk index (HRI), was calculated using the equation below:

$$\text{Total Hazard index (THI) or Health Risk Index (HRI)} = HQ_i \text{ of Nitrate} + HQ_i \text{ of Iron} + HQ_i \text{ of Zinc} \quad (3)$$

If the values of HQ, HRI, or THI are less than one, the water is considered safe for drinking; if the values exceed one, the water is considered unsafe and may pose significant adverse health effects.

2.4. Statistical Analysis

The data were processed using Microsoft Office Excel 2010 to calculate the mean, standard deviation (SD), and HRI.

3. Results and Discussion

The groundwater quality of the selected sites (Figure 1) for the selected parameters, i.e., pH, TDS, TH, NO₃⁻, As, Fe, Cu, Zn, Pb, and Cd, was analysed once every three months, and then the mean value of all twenty-four observations of each site was tabulated in Table 1.

3.1. pH

The pH scale, which ranges from 0 to 14, is a unitless numerical indicator of the acidity or alkalinity of water. The pH of groundwater regulates the ongoing geochemical and physicochemical processes within it (Saalidong et al., 2022). Both human and soil health are affected by pH values outside the neutral range. Acidic groundwater can lead to neurological and cardiovascular disorders in humans because low pH promotes the leaching of toxic metals into the water. The mean pH values in the current study varied from 7.43 to 7.62, suggesting that the water was almost neutral. GWS7 had the lowest pH value (7.43±0.04) and GWS10 the highest (7.62±0.21), with an overall mean of 7.49±0.06. The pH levels were found to be within the BIS guideline (2012) of 6.5 to 8.5 (Figure 2). Singh et al. (2014) reported a similar pH range (7.10-7.85) in various areas of the Baghpat district.

3.2. Total dissolved solids (TDS)

TDS refers to the concentration of dissolved ions and salts in a unit volume of water. The current study found that the mean TDS was 927.58 mg/l±488.97, with the lowest value at GWS6 (514.50 mg/l±12.01) and the highest at GWS8 (1967.50 mg/l±18.01). At every site, the TDS readings were higher than the BIS standard limit of 500 mg/l (Figure 3). TDS levels exceeding the permissible threshold can cause several health issues, including skin rashes, nausea, vomiting, and respiratory irritation (Prasun and Singh, 2025).



Table 1. Physicochemical and heavy metal characteristics of the groundwater collected from the different selected sites.

Parameters / Site	GWS1	GWS2	GWS3	GWS4	GWS5	GWS6	GWS7	GWS8	GWS9	GWS10	Avg. $\pm$ SD
pH	7.44 $\pm$ 0.07	7.48 $\pm$ 0.05	7.46 $\pm$ 0.04	7.49 $\pm$ 0.08	7.54 $\pm$ 0.11	7.48 $\pm$ 0.07	7.43 $\pm$ 0.04	7.56 $\pm$ 0.11	7.43 $\pm$ 0.03	7.62 $\pm$ 0.21	7.49 $\pm$ 0.06
Range	7.29–7.49	7.42–7.56	7.39–7.52	7.35–7.58	7.42–7.67	7.43–7.64	7.35–7.48	7.39–7.78	7.38–7.49	7.39–7.96	—
TDS	563.75 $\pm$ 5.52	1627.75 $\pm$ 38.31	976.63 $\pm$ 22.09	830.25 $\pm$ 33.53	843.38 $\pm$ 5.95	514.50 $\pm$ 12.01	590.38 $\pm$ 7.13	1967.50 $\pm$ 18.01	574.38 $\pm$ 5.93	787.25 $\pm$ 10.59	927.58 $\pm$ 488.97
Range	553.00–570.00	1567.00–1673.00	945.00–998.00	798.00–904.00	837.00–854.00	497.00–534.00	583.00–603.00	1943.00–1990.00	568.00–587.00	773.00–803.00	—
Total Hardness (TH)	311.14 $\pm$ 8.01	831.43 $\pm$ 20.32	412.52 $\pm$ 9.83	306.47 $\pm$ 6.60	490.02 $\pm$ 4.20	229.18 $\pm$ 4.16	318.63 $\pm$ 4.75	745.01 $\pm$ 4.68	431.44 $\pm$ 5.87	415.99 $\pm$ 7.93	449.18 $\pm$ 195.20
Range	298.23–321.34	789.67–852.56	398.56–429.23	298.46–314.48	483.57–495.40	223.67–235.69	311.74–325.12	741.87–755.21	421.97–442.12	408.23–431.76	—
Nitrate (NO ₃ [−])	4.02 $\pm$ 0.97	9.31 $\pm$ 0.39	8.47 $\pm$ 0.51	4.60 $\pm$ 0.29	3.83 $\pm$ 0.33	4.16 $\pm$ 0.11	5.82 $\pm$ 0.30	11.38 $\pm$ 0.92	4.86 $\pm$ 0.18	5.45 $\pm$ 0.19	6.19 $\pm$ 2.61
Range	3.00–5.40	8.56–9.69	7.45–8.98	4.15–4.98	3.08–4.12	3.98–4.34	5.45–6.45	10.13–12.62	4.70–5.12	5.05–5.64	—
Arsenic (As)	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.00
Range	—	—	—	—	—	—	—	—	—	—	—
Iron (Fe)	0.61 $\pm$ 0.08	1.75 $\pm$ 0.31	1.41 $\pm$ 0.08	0.62 $\pm$ 0.06	0.37 $\pm$ 0.04	0.23 $\pm$ 0.04	1.13 $\pm$ 0.10	0.16 $\pm$ 0.03	1.59 $\pm$ 0.22	0.35 $\pm$ 0.03	0.82 $\pm$ 0.60
Range	0.46–0.69	1.09–1.98	1.23–1.49	0.53–0.69	0.32–0.42	0.19–0.29	1.02–1.34	0.12–0.19	1.12–1.78	0.31–0.38	—
Copper (Cu)	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.00
Range	—	—	—	—	—	—	—	—	—	—	—
Zinc (Zn)	0.09 $\pm$ 0.03	0.10 $\pm$ 0.06	0.14 $\pm$ 0.03	0.16 $\pm$ 0.01	1.44 $\pm$ 0.30	0.70 $\pm$ 0.64	0.53 $\pm$ 0.10	0.46 $\pm$ 0.31	0.79 $\pm$ 0.14	0.16 $\pm$ 0.07	0.46 $\pm$ 0.43
Range	0.02–0.12	0.05–0.23	0.11–0.18	0.14–0.18	1.02–1.78	0.12–1.65	0.38–0.67	0.19–0.89	0.67–1.08	0.07–0.27	—
Lead (Pb)	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.00
Range	—	—	—	—	—	—	—	—	—	—	—
Cadmium (Cd)	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	BDL	0.00
Range	—	—	—	—	—	—	—	—	—	—	—

Avg. = Average; SD = Standard Deviation; BDL = Below Detectable Limit. All values are in mg/L except pH (unitless).

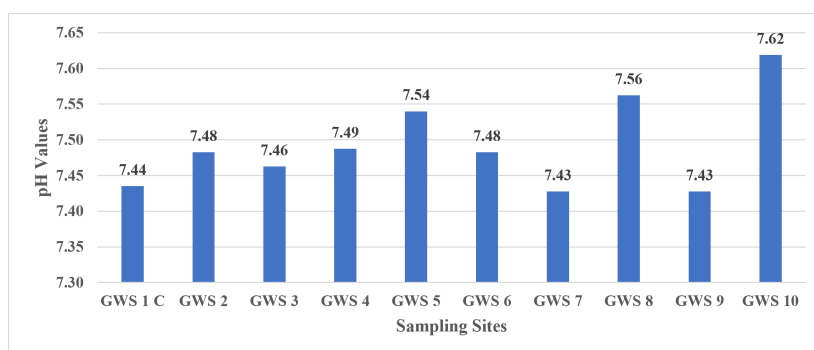


Figure 2. Variation in the mean values of groundwater pH at all the selected sites.

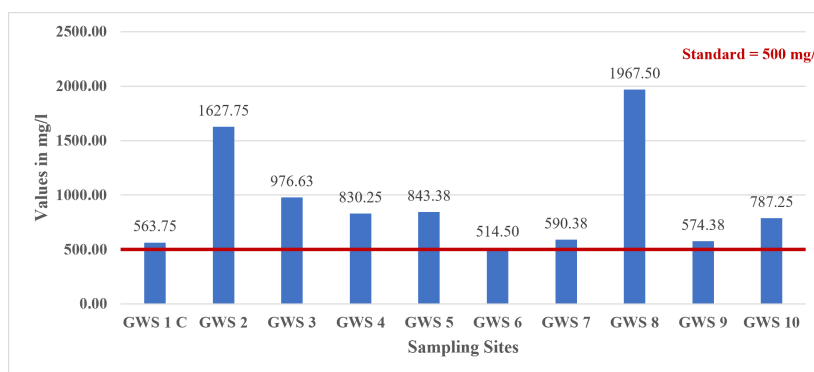


Figure 3. Variation in the mean values of groundwater TDS at all the selected sites.

3.3. Total Hardness (TH)

TH indicates the presence of calcium and magnesium carbonate, chloride, sulphate, and bicarbonate salts in water. While chloride and sulfate contribute to permanent hardness, which requires special water treatment, carbonate and bicarbonate contribute to temporary hardness, which can be eliminated by boiling the water (Islam et al., 2025). Out of all the sites examined, GWS6 had the lowest TH (229.18 mg/l $\pm$ 4.16) and GWS2 the highest (831.43 mg/l $\pm$ 20.32), with a mean TH of 449.18 mg/l $\pm$ 195.20. At every location, TH values exceeded the BIS standard limit of 200 mg/l (Figure 4). High TH levels in drinking water can cause abdominal pain, kidney stone formation, and scaling in water pipes (Li et al., 2022; 2024; He et al., 2026). The elevated levels of TDS and TH in the groundwater of the study area may be attributed to mineral leaching from agricultural operations and waste disposal sites (Prasun and Singh, 2025).

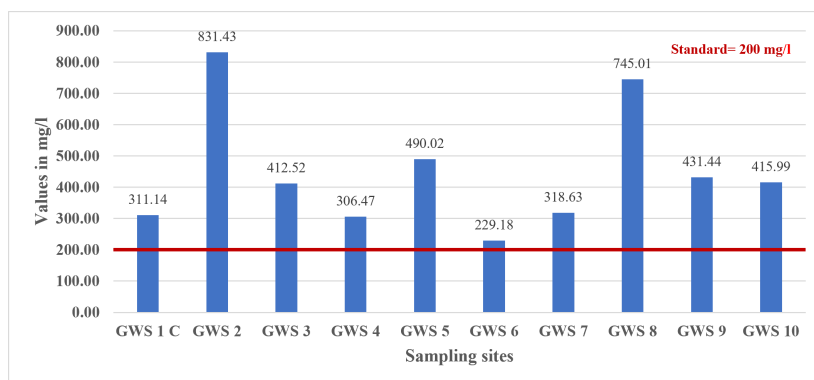


Figure 4. Variation in the mean values of groundwater TH at all the selected sites.

3.4. Nitrate (NO_3^-)

Excessive use of nitrogen-based agrochemicals, untreated or partially treated wastewater discharge, and animal waste disposal are the causes of elevated NO_3^- levels in groundwater (Arman et al., 2024; Sanjupriya et al., 2025; Deep et al., 2025; Ruhela et al., 2026). Elevated NO_3^- concentrations pose a risk to both human health and aquatic ecosystems (Chaurasia et al., 2021; Golaki et al., 2022). Among the total sites examined, GWS1 had the lowest nitrate value ($4.02 \text{ mg/l} \pm 0.97$) while GWS8 had the highest ($11.38 \text{ mg/l} \pm 0.92$), and the average nitrate value was $6.19 \text{ mg/l} \pm 2.61$. At every site, the nitrate levels were below the BIS standard limit of 45 mg/l (Figure 5).

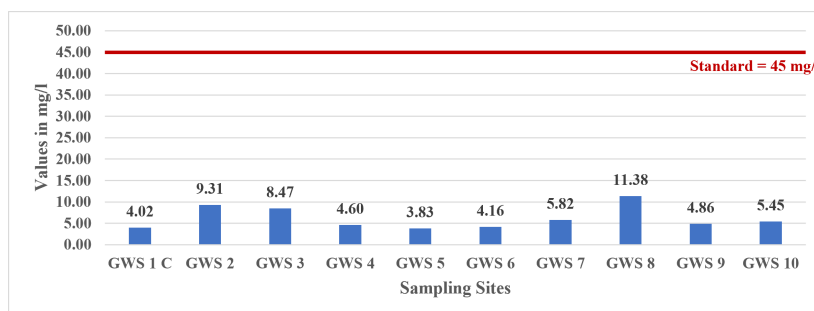


Figure 5. Variation in the mean values of groundwater Nitrate at all the selected sites.

3.5. Iron (Fe)

The majority of Fe in groundwater comes from natural processes including soil erosion and rock weathering. Other anthropogenic factors that contribute iron to groundwater include mining operations and the iron processing industry (Ram et al., 2021). Fe occurs in groundwater predominantly as ferrous ions (Fe^{2+}), which oxidize to ferric ions (Fe^{3+}) on contact with the atmosphere. Of all the sites examined, GWS8 had the lowest Fe value ($0.16 \text{ mg/l} \pm 0.03$), while GWS2 had the highest ($1.75 \text{ mg/l} \pm 0.31$). The average Fe value was $0.82 \text{ mg/l} \pm 0.60$. With the exception of GWS6 and GWS8, all the sites (80% of the samples) had Fe values higher than the BIS permissible limit of 0.3 mg/l (Figure 6). Soil erosion, agricultural runoff, and leaching may be the primary causes of the elevated Fe concentrations in the groundwater of the study area, which is dominated by agriculture and contains areas of sand mining (Narayanan et al., 2025).

3.6. Zinc (Zn)

Zn is an essential micronutrient required for normal growth and development of the human body, although excessive intake is detrimental to human health (Adimalla, 2020a). Both natural and man-made sources, such as rock weathering, agricultural runoff, mining operations, and wastewater disposal, add zinc to groundwater (Gautam et al., 2016). Zn levels ranged from $0.09 \text{ mg/l} \pm 0.03$ at GWS1 to $1.44 \text{ mg/l} \pm 0.30$ at GWS5, with a mean of $0.46 \text{ mg/l} \pm 0.43$. At every site, the Zn values were below the BIS permissible limit of 5.0 mg/l (Figure 7).

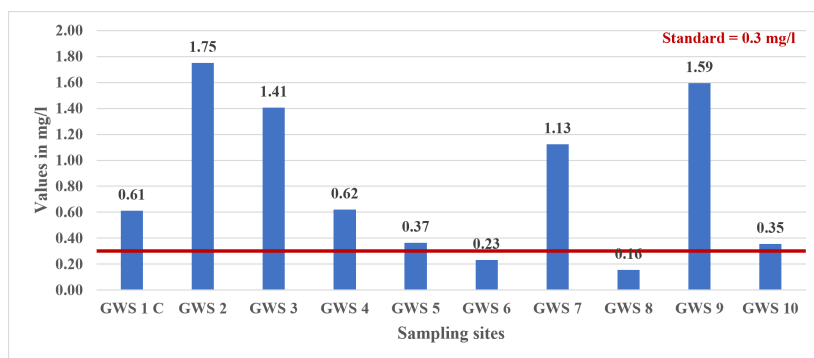


Figure 6. Variation in the mean values of groundwater Iron at all the selected sites.

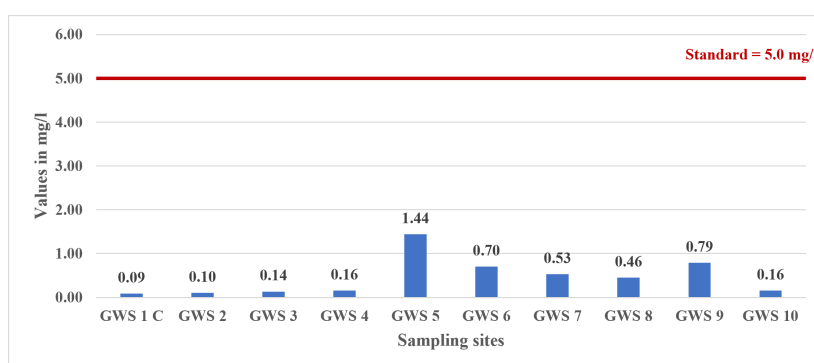


Figure 7. Variation in the mean values of groundwater Zinc at all the selected sites.

3.7. Arsenic (As)

Arsenic (As) is one of the most toxic metalloids encountered in groundwater and enters the aquifer system largely through natural processes such as the weathering of arsenic-bearing sulphide minerals, reductive dissolution of iron oxyhydroxides, and desorption under alkaline conditions. Anthropogenic contributions arise from the application of arsenical pesticides, phosphate fertilizers, coal combustion residues, and industrial effluents. Chronic ingestion of arsenic-rich drinking water is associated with arsenicosis, hyperkeratosis, pigmentation disorders, peripheral vascular disease, and carcinoma of the skin, bladder, and lung, and the element is therefore classified as a Group 1 human carcinogen. Elevated arsenic concentrations have been widely reported from the alluvial aquifers of the middle and lower Ganga plain, where reducing conditions and high organic loading favour arsenic mobilization. In the present investigation, however, the value of arsenic (As) was found below the detectable limit (BDL) at all ten sampling sites throughout the study period, and consequently remained well within the BIS standard permissible limit of 0.01 mg/l. The absence of detectable arsenic suggests that the shallow aquifer of the Baghpat area is dominantly oxidizing in nature and that the local sediments are not enriched in arsenic-bearing minerals. Arsenic was accordingly excluded from the subsequent health risk index computation.

3.8. Lead (Pb)

Lead (Pb) is a non-essential, cumulative toxicant that reaches groundwater through both geogenic and anthropogenic pathways. Natural sources include the weathering of galena and other lead-bearing sulphide minerals, while anthropogenic inputs are dominated by lead-acid battery recycling units, paint and pigment industries, electroplating effluents, vehicular emissions deposited on soils, sewage sludge application, and the corrosion of lead-jointed distribution pipes. Lead has no known beneficial physiological role, and chronic exposure through drinking water is linked to impaired neurodevelopment and reduced intelligence quotient in children, anaemia, hypertension, nephropathy, and reproductive dysfunction in adults. Because young children absorb a substantially larger proportion of ingested lead than adults, they represent the most vulnerable receptor group. In the present study, the value of lead (Pb) was found below the detectable limit (BDL) at all the selected groundwater sampling sites throughout the study period and therefore remained below the BIS standard permissible limit of 0.01 mg/l. This indicates that neither the aquifer sediments nor the surrounding land-use practices are contributing appreciable quantities of lead to the shallow groundwater of



the study area, despite the presence of agricultural and small-scale industrial activity. Lead was consequently not included in the health risk index calculation.

3.9. Cadmium (Cd)

Cadmium (Cd) is a highly mobile and toxic trace metal that occurs naturally in association with zinc, lead, and copper sulphide ores and is released to groundwater through the weathering of these minerals. Anthropogenic sources are generally more significant and include phosphatic fertilizers, electroplating and pigment industries, nickel-cadmium battery waste, sewage sludge, and the leachate generated from unlined municipal solid waste dumping sites. Cadmium is readily taken up by crops and accumulates in the human kidney, where its biological half-life may exceed twenty years. Chronic ingestion is associated with renal tubular dysfunction, proteinuria, osteomalacia and osteoporosis, hypertension, and it has been classified as a probable human carcinogen. In the present investigation, the value of cadmium (Cd) was found below the detectable limit (BDL) at all ten groundwater sampling sites throughout the study period, and thus remained far below the BIS standard permissible limit of 0.003 mg/l. The absence of detectable cadmium indicates that the intensive agricultural fertilizer application in the Baghpat area has not yet resulted in measurable cadmium enrichment of the shallow aquifer, and that no significant industrial cadmium source is present in the vicinity of the sampled wells. Cadmium was therefore excluded from the health risk index computation.

3.10. Copper (Cu)

Copper (Cu) is an essential micronutrient required for haemoglobin synthesis, enzymatic activity, and connective tissue formation, but it becomes toxic when ingested in excess over prolonged periods. Copper enters groundwater naturally through the weathering of chalcopyrite, malachite, and other copper-bearing minerals, and anthropogenically through copper-based fungicides and pesticides used in agriculture, electroplating and wire-drawing effluents, brass and bronze manufacturing waste, and the corrosion of copper plumbing in distribution networks. Excessive intake of copper through drinking water may cause gastrointestinal irritation, nausea and vomiting, and in chronic cases hepatic and renal damage, particularly in infants and in individuals with Wilson's disease or other genetic disorders of copper metabolism. In the present study, the value of copper (Cu) was found below the detectable limit (BDL) at all the selected groundwater sampling sites throughout the study period, and therefore remained well within the BIS standard permissible limit of 0.05 mg/l. This suggests that the aquifer material of the study area is not enriched in copper-bearing minerals and that agricultural fungicide application in the region has not produced detectable copper leaching to the shallow groundwater. Copper was accordingly not considered further in the health risk index calculation.

3.11. Health Risk Index (HRI)

Health risk assessment is an effective tool for quantifying risks from ingesting hazardous substances through drinking water (Sun et al., 2023). Both carcinogenic and non-carcinogenic risks can be assessed based on HRI values (Khan et al., 2023). In the current study, we analysed the groundwater for As, Fe, Cu, Zn, Pb, and Cd. The results revealed BDL values of As, Cu, Pb, and Cd. None of these metals therefore poses a health risk to human beings when their concentrations are compared with BIS standards (Brindha et al., 2020). Therefore, in the current study, Fe and Zn were considered for HRI calculation. Besides this, the values of NO_3^- were also considered.

In the current study, non-carcinogenic health risks due to ingestion were considered. HQ, HRI, and THI values below 1 indicate that the water is safe for drinking, whereas values above 1 indicate that it is unsafe. The obtained results of the HQ of Fe, Zn, and NO_3^- and the cumulative THI are given in Table 2. The THI values at all the selected sites were below 1 for men, women, and children. The values of THI for men ranged from 0.14 to 0.34, for women from 0.16 to 0.39, and for children from 0.16 to 0.43. The values of HQ of nitrate for men ranged from 0.09 to 0.27, for women from 0.11 to 0.32, and for children from 0.12 to 0.37. The values of HQ of iron for men ranged from 0.01 to 0.10, for women from 0.01 to 0.10, and for children from 0.01 to 0.13. The values of HQ of zinc for men ranged from 0.01 to 0.18, for women from 0.01 to 0.19, and for children from 0.00 to 0.02 (Table 2).

Among the selected parameters, i.e., NO_3^- , Fe, and Zn, the highest risk contribution was observed for NO_3^- , followed by Fe and Zn. Among men, women, and children, the highest risk was observed for children, followed by women and men. A similar trend was also observed by Adimalla and Wang (2018), Zhaoyong et al. (2019), Adimalla (2020b), Khan et al. (2023), Sun et al. (2023), and Ahamad et al. (2024). The higher health risks to children may be attributed to their lower body weight and correspondingly higher water intake per unit body weight (Biswas et al., 2024). Jeelani et al. (2025) carried out a health risk assessment in two selected villages of Ghaziabad (Sharfuddinpur Javli) and Baghpat district (Gangnauli) using a mixed-method approach comprising interviews, social mapping, focus group discussions, community interaction, and a literature survey. The authors found that the degraded environmental quality led to a substantial burden of disease. Goyal et al. (2022) analysed the sludge and water of ponds located in Baghpat district and observed concentrations of the studied metals beyond the permissible limits. The authors also reported high health risks to human beings from these metals (HRI = 0.831 to 21.76).



Table 2. Hazard quotient (HQ) and total hazard index (THI) of non-carcinogenic risk for men, women, and children due to nitrate, iron, and zinc ingestion of groundwater.

Sites	HQ* of Nitrate			HQ _i of Iron			HQ _i of Zinc			THI		
	Men	Women	Child	Men	Women	Child	Men	Women	Child	Men	Women	Child
GWS1	0.10	0.11	0.13	0.03	0.04	0.05	0.01	0.01	0.00	0.14	0.16	0.18
GWS2	0.22	0.26	0.30	0.10	0.10	0.13	0.01	0.01	0.00	0.33	0.38	0.43
GWS3	0.20	0.24	0.28	0.08	0.08	0.10	0.02	0.02	0.00	0.30	0.34	0.38
GWS4	0.11	0.13	0.15	0.03	0.04	0.05	0.02	0.02	0.00	0.16	0.19	0.20
GWS5	0.09	0.11	0.12	0.02	0.02	0.03	0.18	0.19	0.02	0.30	0.32	0.18
GWS6	0.10	0.12	0.14	0.01	0.01	0.02	0.09	0.09	0.01	0.20	0.23	0.16
GWS7	0.14	0.17	0.19	0.06	0.06	0.08	0.07	0.07	0.01	0.27	0.30	0.28
GWS8	0.27	0.32	0.37	0.01	0.01	0.01	0.06	0.06	0.01	0.34	0.39	0.39
GWS9	0.12	0.14	0.16	0.09	0.09	0.12	0.10	0.11	0.01	0.31	0.34	0.29
GWS10	0.13	0.15	0.18	0.02	0.02	0.03	0.02	0.02	0.00	0.17	0.20	0.21

4. Conclusion

The present study provides a comprehensive assessment of groundwater quality and associated non-carcinogenic health risks in the agriculturally intensive Baghpat city of western Uttar Pradesh, India. The elevated concentrations of TDS and TH observed in the groundwater samples suggest significant mineral dissolution and geochemical interactions within the aquifer system. These elevated values may also be influenced by agricultural activities, including the long-term application of fertilizers and other agrochemicals, as well as the possible leaching of contaminants from improper waste disposal sites. Although nitrate concentrations were found to be within the permissible limits prescribed by the BIS, their higher concentrations relative to those reported in surrounding areas such as Meerut indicate the potential influence of agricultural practices and emphasize the need for continued monitoring. Among the analysed heavy metals, As, Cd, Pb, and Cu were found below the detectable limits, whereas Fe and Zn were detected in the groundwater samples. Zn concentrations remained within the BIS permissible limit, indicating no significant concern related to Zn contamination. In contrast, Fe concentrations exceeded the permissible limit of 0.3 mg/l at most sampling locations, except GWS6 and GWS8. The elevated Fe concentration may be attributed to natural geogenic processes, including the weathering and dissolution of Fe-bearing minerals, as well as possible soil erosion and local environmental conditions. Therefore, further hydrogeochemical investigations are recommended to distinguish between natural and anthropogenic sources of iron contamination. The health risk assessment showed that the non-carcinogenic HRI values for men, women, and children were below the critical threshold of 1, indicating that groundwater consumption does not currently pose significant non-carcinogenic health risks. However, the relative contribution of individual contaminants showed that nitrate was the major contributor to the overall health risk, followed by iron and zinc. Among the exposed population groups, children were found to be the most vulnerable, followed by women and men, highlighting the importance of considering age- and gender-specific exposure characteristics in groundwater health risk assessments. Overall, the study indicates that groundwater in the investigated area is generally safe with respect to the detected heavy metals and associated non-carcinogenic health risks. However, the elevated levels of TDS, TH, and iron at several locations may adversely affect the suitability of groundwater for direct drinking purposes and necessitate appropriate treatment before consumption. Long-term monitoring and integrated groundwater management strategies are recommended to protect groundwater resources and ensure a safe and sustainable drinking water supply for the local population.

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